

Tribological performances and self-lubricating mechanism of monomer casting nylon-6 composite coatings containing lube base oil-loaded microcapsules

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ABSTRACT

The tribological properties of monomer casting nylon-6 (MC PA6) play an important role in ensuring energy consumption, component integrity and reliability in construction machinery. In this work, MC PA6 coatings containing different amount of lube base oil-loaded poly(urea-formaldehyde) microcapsules are prepared, and the tribological properties of the composite coatings are investigated. Compared with the pure coating, the friction coefficient and specific wear rate of the composite coating containing 20 wt% of oil-loaded microcapsules are 0.16 and $0.47 \times 10^{-5} \text{ mm}^3/\text{Nm}$, which correspond to decreases of 56.7% and 91.8% respectively. Moreover, by introducing the oil-loaded microcapsules, the heat produced during friction is greatly reduced and the friction temperature of all composite coatings ($\sim 40\text{--}50^\circ\text{C}$) decreases more than 40°C , as compared with that of pure coating (93.7°C). Self-lubricating mechanism is investigated through microscopy and EDS analysis, which indicate that the improvement of tribological properties is attributed to the formation of uniform, continuous and thin lubricating film on the worn surface. This study would provide significant guidance for the design and preparation of MC PA6 composite coatings with high self-lubricating properties.

1. Introduction

Monomer casting nylon-6 (MC PA6), synthesized by anionic ring-opening polymerization of caprolactam, is one of the most commercially prominent engineering plastics [1]. Attributed to its high molecular weight, high crystallinity, excellent mechanical properties, low preparation cost, high reaction rate and short production cycle, MC PA6 has been widely investigated and applied in automotive, shipbuilding and other engineering fields as an alternative material for gears, bearings, sliders and other metal parts [2,3]. However, MC PA6 has high wear rate and high friction coefficient, which causes energy consumption, material loss, parts failure and reliability reduction under harsh conditions (high load, high speed and long-time wear process, etc.), limiting the application of MC PA6 in construction machinery [4].

Many attempts have been made to improve the tribological

properties of polymer materials in response to the challenges associated with harsh friction conditions, among which, two strategies are commonly adopted, which are adding solid lubricants and liquid lubricants. For MC PA6, solid lubricants with low interlamellar friction coefficient such as graphite [5], graphene nanosheets [4,6], boron nitride [7] and polytetrafluoroethylene [8] are usually employed, which play an adhesive role in reducing the wear rate of polymer composites by interlayer sliding, and forming thin and uniform transfer films. In addition, solid lubricants like glass fibers [9,10] are introduced in MC PA6 and act as load-bearing to enhance the mechanical properties of the matrix so as to improve the anti-wear property of the composite [11]. Unfortunately, most of these solid lubricants are prone to agglomerate because of their high surface energy, and thus are difficult to disperse and process, resulting in relatively expensive processes [12]. Besides, solid lubricants fall off easily during the friction processes, making it

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extremely unfavorable for their anti-friction and anti-wear performances [13].

Adding liquid lubricants such as ionic liquid [14], white oil [5], lithium grease [15], and silicone oil [16] could provide good tribological performance for polymers, due to the formation of oil film [17]. However, directly introducing liquid lubricants into polymers suffers from dispersion difficulties and sharp deterioration of mechanical properties [18], while employing liquid lubricants for external lubrication is not suitable for oil-free applications and is also not environmental friendly [19]. In view of this, some studies have proposed to solve these problems by adding microcapsules containing liquid lubricants into polymers [20]. When the polymer surface is rubbed, the microcapsules get ruptured and release the encapsulated lubricants onto the rubbing surface, forming a boundary lubrication film that significantly improves the tribological properties [21]. Generally, materials especially polysulfone [22], poly(urea-formaldehyde) (PUF) [21], poly(melamine-formaldehyde) [23] and polyurethane [24] are adopted as the shell materials to encapsulate liquid lubricants like ionic liquids [14,25], paraffin wax [26], and lubricant oil [23], etc. to synthesize microcapsules. These oil-loaded microcapsules have been added into polymer materials such as epoxy [23,27] and polypropylene [22], etc. to investigate their self-lubricating properties, obtaining remarkable results.

PUF is commonly accepted as a good candidate for encapsulating lubricants through in-situ polymerization technology, due to their excellent comprehensive properties including high mechanical strength, good sealing performance, strong chemical resistance, high encapsulation rate and low cost in industrial scale, etc. [28]. Besides, the rough surfaces of PUF microcapsules are also beneficial to the formation of mechanical interlocks with the polymer matrix, inhibiting the falling off of the microcapsules [29]. On the other hand, lube base oils, composed of saturated hydrocarbons with high content and few impurities, bear good thermal stability, low-temperature fluidity and excellent lubrication performance, and thus are widely applied in various industrial lubrication fields as lubricants. Nonetheless, to our best knowledge, introducing PUF microcapsules containing lube base oil into MC PA6 coatings and systematically investigating the tribological performance of the resulting composites have not been reported. We believe that introducing oil-loaded microcapsules into MC PA6 will create a composite coating with high self-lubricating performance that can be applied in a wide range of applications. Besides, the self-lubricating mechanism is of great significance, which also needs exploration and clarification imminently.

Therefore, in this work, PUF microcapsules containing lube base oil synthesized by in-situ polymerization are successfully introduced into MC PA6 to investigate the effect of oil-loaded microcapsules on the tribological behavior of MC PA6 composite coatings. We detail below the systematic study of friction coefficient, specific wear rate, and the hardness of the composite coatings with different microcapsule contents, the characterizations of the worn surfaces, the friction temperature during tribological tests as well as the self-lubricating mechanism. This study would provide significant guidance for the design and preparation of MC PA6 composite coatings with high self-lubricating properties that can even be applied in harsh lubrication conditions.

2. Experimental section

2.1. Materials

32# Lube base oil (kinematics viscosity at 40 °C: 31.17 mm²/s) was purchased from Jiarun Petrochemical Co. Ltd. (Wuxi, China). Urea, formaldehyde aqueous solution (37 wt%), n-octanol, ammonium chloride (NH₄Cl), hydrochloric acid (HCl), triethanolamine (TEA), anhydrous ethanol, isopropanol, caprolactam (CL) monomer, sodium hydroxide (NaOH) as the main catalyst, and formic acid aqueous solution (88 wt%) were obtained in analytical grade from Kelong Chemical Reagents Co. Ltd. (Chengdu, China). Sodium lignosulfonate with a low

weight-average molecular weight of 3100 g/mol was purchased from Nanjing Shanhu Chemical Co. Ltd. (Nanjing, China). Resorcinol and Hexamethylene diisocyanate (HDI) as the auxiliary catalyst, were provided by Aladdin. Distilled water was used in the study. All chemicals were used as received without any further purification.

2.2. Preparation of PUF microcapsules containing lube base oil

The PUF microcapsules containing lube base oil were prepared by two-step in-situ polymerization method, which gives microcapsules with rougher shell structures [30] beneficial to the formation of mechanical interlocks with polymer matrix [29]. Based on that PUF microcapsule containing 32# lube base oil has the best comprehensive properties [30], and the subsequent composite coatings have the best tribological properties (Please see the Supporting Information for details), 32# lube base oil was particularly adopted as the core material. Briefly, 2.5 g urea was added into 5 g formaldehyde aqueous solution (37 wt%) under vigorous stirring, followed by adding TEA to adjust the pH to 8–9 and then heating the mixture at 65 °C for 1 h to acquire the UF pre-polymer (pre-UF). 0.25 g Resorcinol, 0.25 g NH₄Cl, and 0.2 wt% sodium lignosulfonate aqueous solutions were mixed well in 150 mL deionized water to form a uniform solution. 7.5 g 32# lube base oil was then added into the solution and homogenization shearing was applied using a digital homogenizer (Ultra Turrax T25, IKA, Germany) to allow the emulsification to occur. Thereafter, the pre-UF solution was mixed with the above emulsion, and the pH of the mixture was adjusted to 3–4 by adding HCl (1 mol/L). The resultant mixture was then kept at 35 °C for 30 min and 60 °C for 2 h under stirring. Finally, the suspension of PUF microcapsules was vacuum filtered, rinsed with distilled water, ethanol and isopropanol, respectively, and then dried by freeze drying for 24 h.

2.3. Preparation of MC PA6

Firstly, 140 g caprolactam monomer was put into a three-necked flask and heated to 140 °C. After complete melting, the melt was refluxed under vacuum conditions for 30 min to remove the internal residual moisture. Afterwards, 0.56 g NaOH (0.4 wt%) was added under vigorous stirring. The melt was refluxed under vacuum for another 30 min and 0.64 g HDI (0.46 wt%) was added. After a quick and well mix, the melt was cast into a mold with the size of 150 × 150 × 4 mm pre-heated to 170 °C. The polymerization lasted for 30 min and the sample of MC PA6 were then demolded [31].

2.4. Preparation of MC PA6/PUF microcapsules composite coatings

The synthetic procedure for the MC PA6/PUF microcapsules composite coatings is schematically shown in Fig. 1. Firstly, MC PA6 was dissolved in formic acid with a concentration of 10 wt% at ambient temperature. Different amount of oil-loaded PUF microcapsules were then dispersed in the MC PA6 solution by 10-min stirring and 30-s sonication to obtain mixtures with 5, 10, 15, 20, 30, 40 wt% microcapsules. The mixtures were continuously stirred at 60 °C for another 1 h to let formic acid evaporate partially (MC PA6 concentration: 15–20 wt %) and then blade-coated on the MC PA6 substrates pre-cut into the size of 60 × 60 × 4 mm to obtain the MC PA6/PUF microcapsules composite coatings. Finally, the resultant composites were dried at 55 °C for 12 h to completely remove the residual formic acid to form microcapsule loaded MC PA6 composite coatings with thicknesses of ~300 μm. We define these composite coatings with the sample codes of C5, C10, C15, C20, C30, C40, respectively, depending on the microcapsules they contained (5, 10, 15, 20, 30, 40 wt%, respectively). For comparison, we also prepared a pure MC PA6 coating with the same method, and define its sample code as C0.

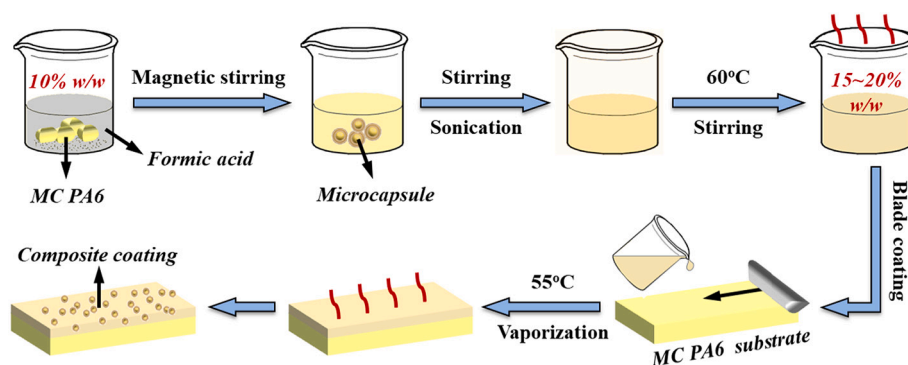


Fig. 1. Schematic diagram of the synthetic route to the fabrication of MC PA6 composite coatings containing oil-loaded PUF microcapsules on a MC PA6 substrate.

2.5. Characterizations

Morphologies of the cryogenically fractured and worn surfaces of MC PA6 composite coatings were observed using a field-emission scanning electron microscope (FE-SEM, Nova Nano SEM 450, FEI, USA). All samples were sputter-coated a thin layer of gold, and observed at an accelerating voltage of 5 kV. In addition, the elemental compositions of the worn surfaces were analyzed via an Energy Dispersive X-ray Spectroscopy (EDS, Utm Max, Oxford Instruments, UK) coupled with the FE-SEM. A laser scanning confocal microscope (LSCM, LSM700, Zeiss, Germany) was adopted to further investigate the worn surfaces of MC PA6 composite coatings.

To evaluate the resistance of composite coatings to mechanical indentation induced localized plastic deformation at ambient temperature, Shore hardness (HD) was measured with a hardness tester (LX-D, Shanghai Chuanlu measure Co., LTD), according to ASTM D2240 ISO/R868. The standard Shore indenter was pressed vertically into the planar specimen, and the reading finished in ~ 1 s after the bottom was in full contact with the specimen. Five valid indentations were measured for each sample at different locations.

In order to measure the tribological performance of MC PA6/microcapsules composite coatings, dry sliding wear tests were carried out on a block-on-ring testing machine (M-200, Beijing Guance Test Instrument CO., LTD). Specimens with the size of $30 \times 7 \times 4$ mm were prepared and rotated against a 45# steel ring for 30 min at a sliding velocity of 0.42 m/s and a pressure of 85 N under ambient condition. Before each test, the surface of steel ring were polished with 1000 grit paper for 3 min, and washed with anhydrous ethanol. At least 3 repeated tests were conducted for each sample and the real-time friction coefficient was recorded online. The time-averaged friction coefficient after the running-in stage (vide infra) was calculated and defined as friction coefficient of the coatings. The specific wear volumes V (mm^3) and specific wear rate W (mm^3/Nm) of the specimens were determined according to Eqs. (2) and (3), respectively:

$$V = B \times \left[\frac{\pi R^2}{180} \times \arcsin\left(\frac{b}{2R}\right) - \frac{b}{2} \sqrt{R^2 - \frac{b^2}{4}} \right] \quad (2)$$

$$W = \frac{V}{F \times L} \quad (3)$$

where B (mm) is the specimens width, R (mm) is the radius of the 45# steel ring (standard value: 20 mm), b (mm) is the width of the wear tracks, F (N) is the load, and L (m) is the sliding distance.

The tribological testing machine M-200 was equipped with an infrared temperature probe, allowing us to record the real-time temperature at the central position of the rotation axis fitted with the steel ring during the test. We also examined the surface temperature of the specimen and the steel counterface using an infrared thermal imager (IR-160 P, RNO, USA) when the test was carried out to 1800 s.

3. Results and discussion

3.1. Cross sectional morphologies of the MC PA6 composite coatings

Fig. 2 shows the cross sections of pure MC PA6 coating (C0) and composite coatings with different contents of oil-loaded PUF microcapsules (C5, C10, C15, C20, C30, C40). Consistent with the blade-coating method, all coatings exhibit an approximately equal thickness of ~ 300 μm . As depicted in Fig. 2a and a₁, the cross section of the pure MC PA6 coating is rather compact, without any recognizable micropores. The cross sections of the microcapsule-loaded composite coatings presented in Fig. 2b–g demonstrate that their internal structures are also very compact, except for the regions where the microcapsules exist. Discrete micropores are observed in Fig. 2b–e, which corresponds to the positions where microcapsules occupy but fall off during cryogenical fracturing. However, local agglomeration of microcapsules in the matrix occurs when excessive content is loaded, as seen from Fig. 2f–g. Additionally, all coatings (C0–C40) bond well with the MC PA6 substrate without any obvious interfaces observed, attributed to the identical material composition of the coating matrix and substrate, and the latter could dissolve slowly in the formic acid at room temperature during the blade coating process to enhance the bonding between the coating and the substrate [32].

3.2. Hardness of the MC PA6 composite coatings

In the process of friction, the hardness of a material is considered an important index because it reflects the ability to resist the pressure generated by friction [33]. The hardness of the MC PA6 coatings containing different contents of microcapsules is presented in Fig. 3. It is quite obvious that the microcapsules-loaded MC PA6 composite coatings all exhibit a decrease in hardness, as compared to the pure coating. Considering the high temperature generated at the surface of pure coating during friction test (~ 100 $^{\circ}\text{C}$, vide infra), which would affect its hardness, we only discuss the hardness of the composite coatings. From Fig. 3 we can see that the hardness of composite coatings containing 5–20 wt% microcapsules is similar, all exceeding ~ 75 HD and with a deviation of < 2 HD. Nevertheless, the hardness of the coatings containing 30 wt% and 40 wt% microcapsules reduces significantly, which are 72 HD (C30) and 64 HD (C40), respectively. This indicates that excessive microcapsules cause dispersion difficulty and agglomeration, which weakens the mechanical interlocking between the microcapsule and the matrix, resulting in a poor cohesion force and internal loose structure of the composite coatings [34,35].

3.3. Tribological performances of the MC PA6 composite coatings

The friction and wear tests of self-lubricating MC PA6 composite coatings containing various amounts of microcapsules are conducted, and the corresponding friction coefficients and specific wear rates are

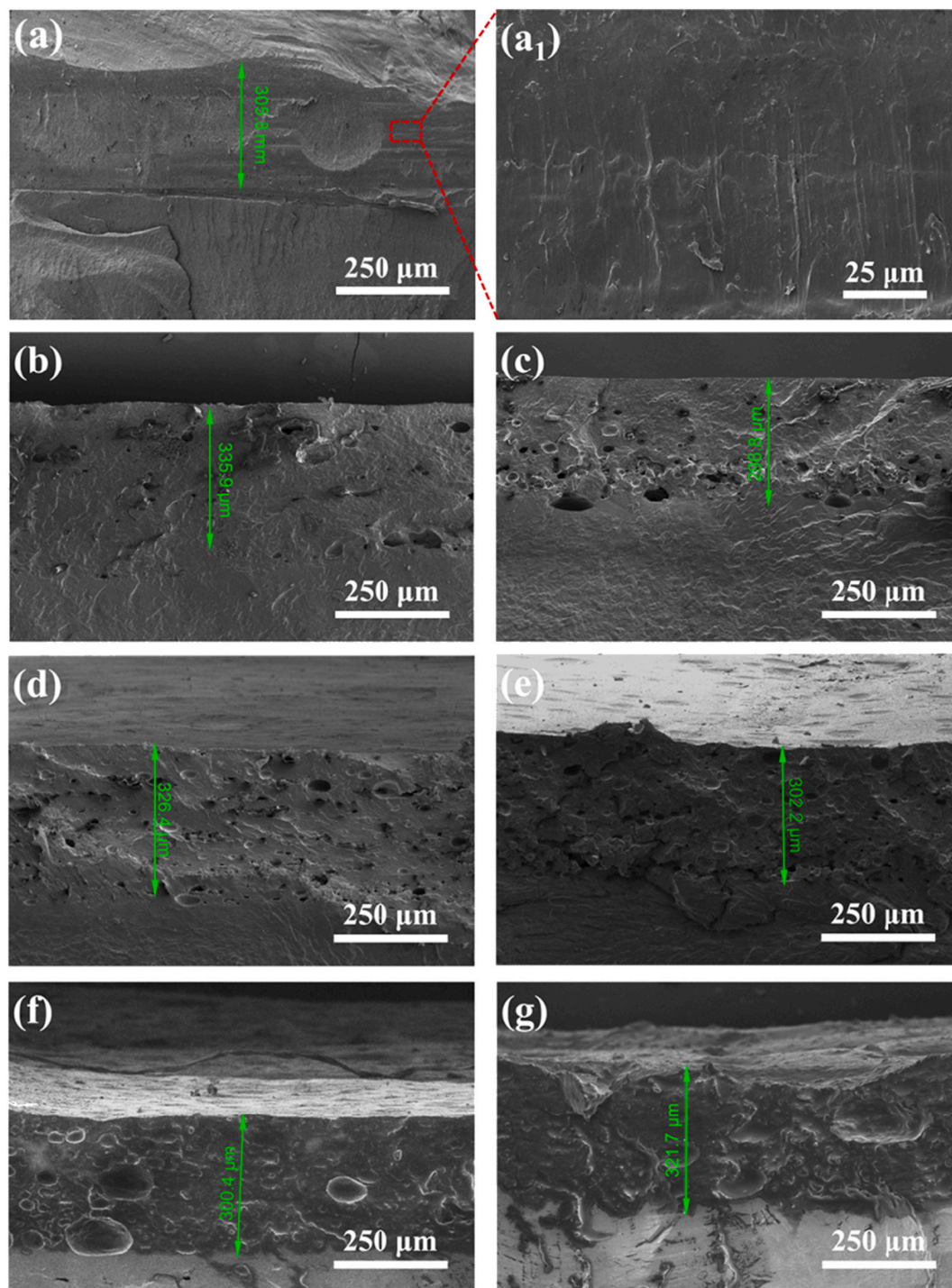


Fig. 2. SEM images of the cross-section of pure MC PA6 and MC PA6 composite coatings with different microcapsule contents. (a, a₁) C0; (b) C5; (c) C10; (d) C15; (e) C20; (f) C30; (g) C40.

shown in Fig. 4. As presented in Fig. 4a, at the beginning of the test (running-in stage), the pure coating shows a dramatic increase in friction coefficient to ~ 0.8 , which then sharply drops and gradually levels off after approximately 500 s. Compared with the pure coating, the friction coefficients of the composite coatings level off much sooner and the running-in stage lasts for only about 50 s for all the composites. The fluctuation of the friction coefficient curves for all samples over time may be due to the repeated formation, adhesion and separation of transfer film during friction and wear tests [36,37]. According to Fig. 4b, the frictional coefficient and specific wear rate of the pure MC PA6

coating are 0.37 and $5.75 \times 10^{-5} \text{ mm}^3/\text{Nm}$, respectively. Noteworthy, by incorporating only 5 wt% microcapsules into MC PA6 coating, the friction coefficient and wear rate dramatically decrease to 0.26 and $1.15 \times 10^{-5} \text{ mm}^3/\text{Nm}$, respectively, indicating that the microcapsules are effective and efficient in improving the tribological properties of MC PA6. Increasing the microcapsule content from 5 wt% to 20 wt%, the frictional coefficient and specific wear rate of the composites decrease monotonically. The composite coating containing 20 wt% oil-loaded microcapsules (C20) obtain a friction coefficient and a specific wear rate of 0.16 and $0.47 \times 10^{-5} \text{ mm}^3/\text{Nm}$ respectively, which are 56.7%

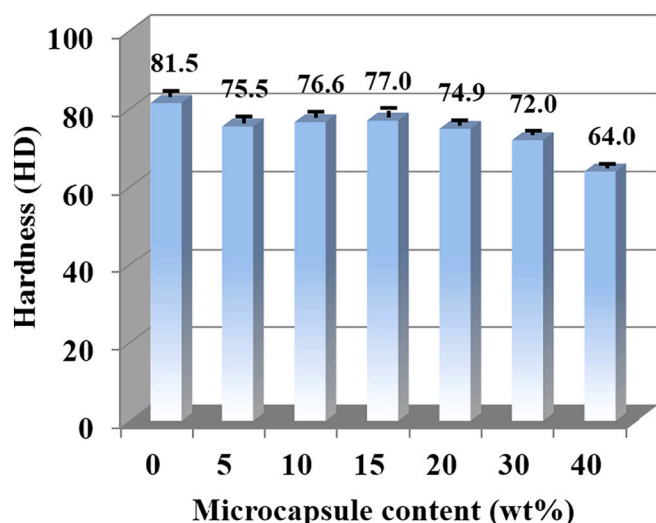


Fig. 3. The hardness of MC PA6 self-lubricating coatings with different microcapsule contents.

and 91.8% lower than that of the pure coating, respectively. Further increasing the microcapsule content, the friction coefficient continues to reduce to 0.15 and 0.14, for C30 and C40, respectively, while the wear rate increases, which might come from the internal loose structure and the decrease of hardness of the composite coatings (*vide supra*). These results demonstrate that introducing a certain amount of oil-loaded PUF microcapsules into MC PA6 coatings could greatly improve the wear resistance under dry sliding condition. Although introducing oil-loaded microcapsules may have slightly lower reduction effect on friction coefficient and wear as compared with directly adding oil [38,39], it does not lead to significant reduction in mechanical properties of MC PA6 as the latter does [18,38].

3.4. Temperature variations of the MC PA6 composite coatings during friction measurements

As is known to all, during the process of friction, the mechanical energy partially transforms into heat energy. Importantly, heat accumulation occurs and temperature increases at the contact region, causing a change in mechanical properties (*i.e.* hardness) of material, which in return affects the tribological behaviors [40]. Fig. 5 presents the detected temperature information of specimens during friction measurements. Surprisingly, introducing microcapsules into MC PA6 causes a dramatic drop in the real-time friction temperature, from nearly 100 °C to less than 45 °C, detected by the built-in temperature probe

(Fig. 5a). Consistent results are obtained from the infrared thermal images (Fig. 5b–h), where the temperature near the contact area for the pure MC PA6 coating reaches ~93.7 °C, while that of composite coatings are only between 40 °C and 50 °C (Table 1).

As discussed earlier, in the first ~500 s of the friction process, the friction coefficient of pure MC PA6 coating rises rapidly to a peak of ~0.8, then drops to a steady-state value of ~0.37 (*vide supra*). This process is accompanied by a sudden increase in temperature to ~100 °C, as shown in Fig. 5b and Table 1, owing to the heat accumulation and the low thermal conductivity of MC PA6 (0.41 W/mK) [41]. The increase in temperature, on one hand, brings a deterioration of elastic modulus of the pure coating, thus the contact area and friction coefficient increase (to ~0.8) [42]. On the other hand, the temperature increase to above the T_g of MC PA6 (56.4 °C [43]) also leads to thermal softening of the worn surface, forming an interfacial layer with low shear strength, which acts as a lubricant instead and contributes to the significant reduction (from ~0.8 to ~0.37) of the friction coefficient [44]. However, it must be pointed out that thermal softening would result in noticeable deterioration of the mechanical properties of pure MC PA6 coating, which reduces the resistance to friction forces and sharply increases the wear loss of the coating [42,45], as also have been observed in Fig. 4b. Therefore, the addition of oil-loaded PUF microcapsules reduces the friction coefficient and wear rate of MC PA6 coating, making an effective control over the heat generation and friction temperature. The reduction in temperature, in return, avoids thermal softening and maintains the intrinsic mechanical properties of the composite coatings, preventing further deterioration of the tribological properties.

3.5. Worn surface morphologies and self-lubricating mechanism of the MC PA6 composite coatings

To investigate the self-lubrication mechanism of the MC PA6 composite coatings, we studied the morphologies of the worn surfaces first. Figs. 6 and 7 show respectively the SEM images and LSCM images of worn surface of pure MC PA6 coating (C0) and composite coatings (C5–C40) after friction test. As indicated by the solid yellow arrows in Fig. 6a₁–a₇, the wear-scar width of MC PA6 coatings decreases first then increases, and reaches the minimum at microcapsule content of 20 wt%. Compared with that of the pure MC PA6 coating, broken microcapsules (indicated by the dashed red circles) and tiny oil droplets (indicated by the dashed yellow arrows) are clearly recognized on the worn surfaces of the composite coatings, as shown in Figs. 6c₁–c₇ and 7. In addition, it is reported that the polymer material tends to transfer to the metallic counter surface when it slides against the metallic grinding ring [46]. The oil droplets released from the ruptured microcapsules and the transferred polymer material combine to form lubricating films (indicated by the white solid arrows in Fig. 6c₃–c₇) at the interface, providing

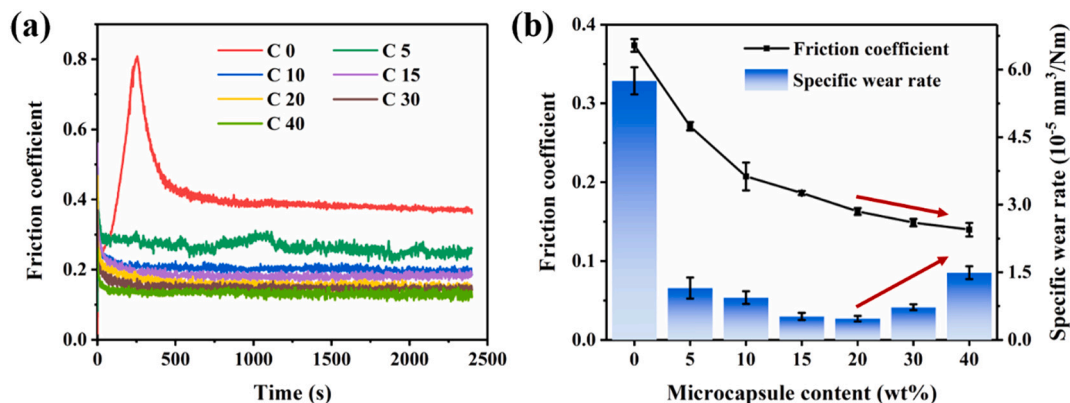


Fig. 4. Tribological properties of MC PA6 composite coatings: (a) the real-time friction coefficient curves of the composite coatings during friction tests (b) friction coefficient and specific wear rate of MC PA6 composite coatings with different microcapsule contents.

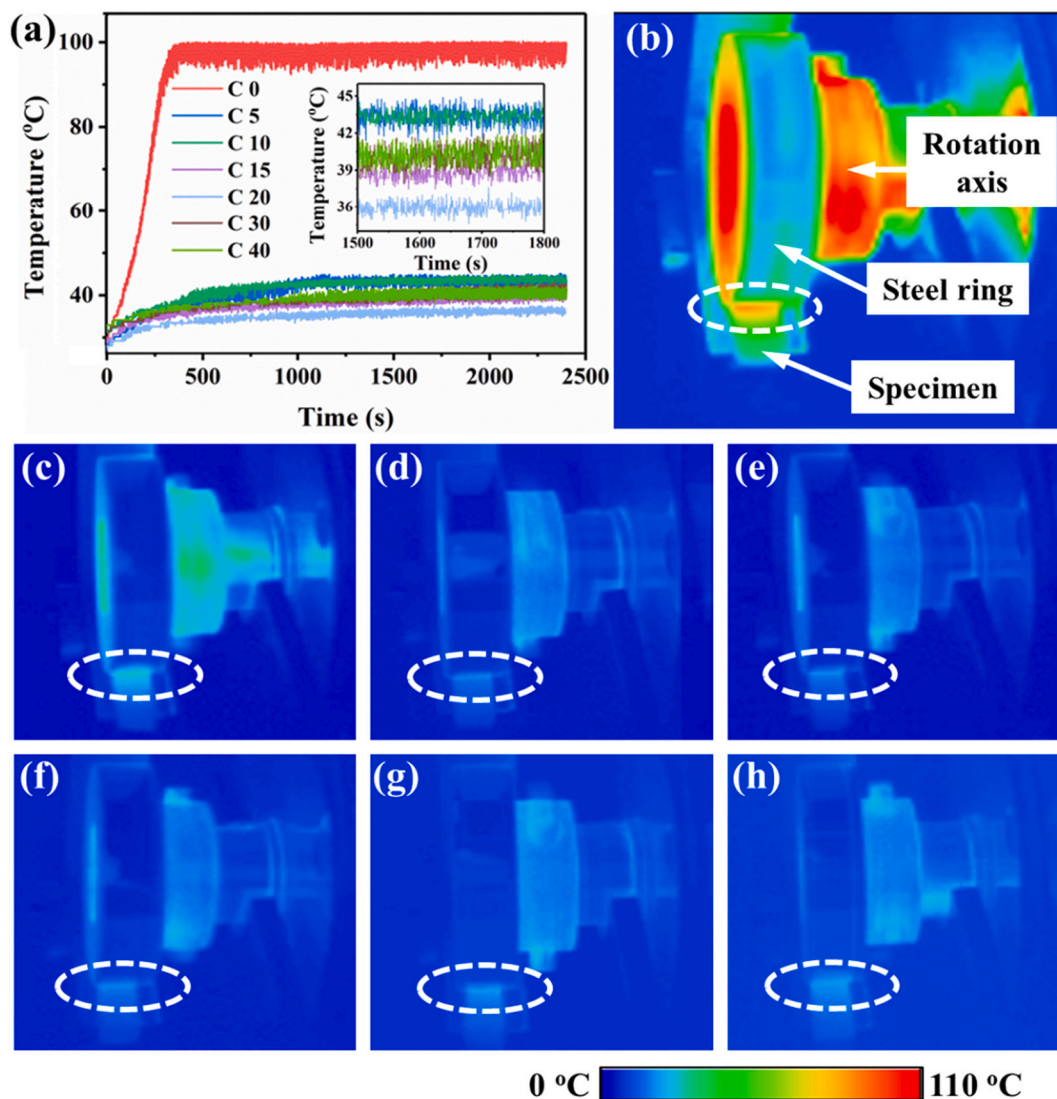


Fig. 5. (a) Real-time temperature of the central position of the rotation axis fitted with the steel ring and (b–h) infrared thermal images of C0, C5, C10, C15, C20, C30, and C40 samples measured at 1800 s of the friction test. The white dashed ellipses in (b–h) indicate the contact areas between the specimens and the steel ring.

Table 1

The temperatures near the contact areas (indicated by the white dashed ellipses in Fig. 5b–h) at 1800 s in the friction measurements.

Sample code	C0	C5	C10	C15	C20	C30	C40
Temperature (°C)	93.7	50.3	42.3	42.0	41.7	43.0	43.8

a low shear resistance layer to protect the matrix, which is quite significant to the tribological behavior of the polymer under dry friction conditions [47,48]. It is obvious that no lubricating film is observed for the pure MC PA6 coating (Figs. 6c₁ and 7a), while the composite coatings containing 5, 10 and 15 wt% of microcapsules (C5, C10 and C15, respectively) exhibit discontinuous thin lubricating films (Figs. 6c₂–c₄ and 7b–d). A uniform, continuous and thin lubricating film is formed on the worn surface of composite coating with 20 wt% microcapsules (C20), as shown in Figs. 6c₅ and 7e. As for the worn surfaces of C30 and C40 composite coatings, uneven thick lubricating films are observed, as illustrated in Fig. 6c₆–c₇, due to the excess loading of oil-containing microcapsules.

To further explore the mechanism of how the microcapsules promote the tribological performance of MC PA6 coating, the worn surfaces of all samples are investigated by EDS, the results of which are shown in

Table 2. We know that the main elements of MC PA6 and PUF shells are C, N, O and H, without other characteristic elements [49,50], while the lube base oil is mainly composed of saturated hydrocarbons with low impurity content (i.e. contain mainly C and H elements). From Table 2 we can see that, compared with the elements on the worn surface of pure MC PA6 coating, there is no significant difference from that of composite coatings with 5, 10, 15, 20 wt% of microcapsules. Nevertheless, It is worth noting that the content of O and N elements on the worn surface of C30 suddenly drop to 11.5 and 5.3 wt%, respectively, while those of C40 are close to 0 wt% (0 and 0.5 wt%), indicating that the surface is predominantly composed of C and H elements. These results further prove the existence of oil films on the surface, which is the main reason for C30 and C40 to have extremely low friction coefficients of 0.15 and 0.14 (vide supra). However, the C30 and C40 show slight increase in wear scar width and specific wear rate than C20 (vide supra), due to the excessive lube base oil which weakens the adhesion strength of lubricating film, and causes easy exfoliation of the formed film [23]. Another reason is that the significant decrease in composite hardness (vide supra) weakens the loadbearing capabilities of worn surfaces, making the composite coatings vulnerable to be damaged when suffering from friction [51,52].

The schematic diagrams of anti-friction and anti-wear mechanism for

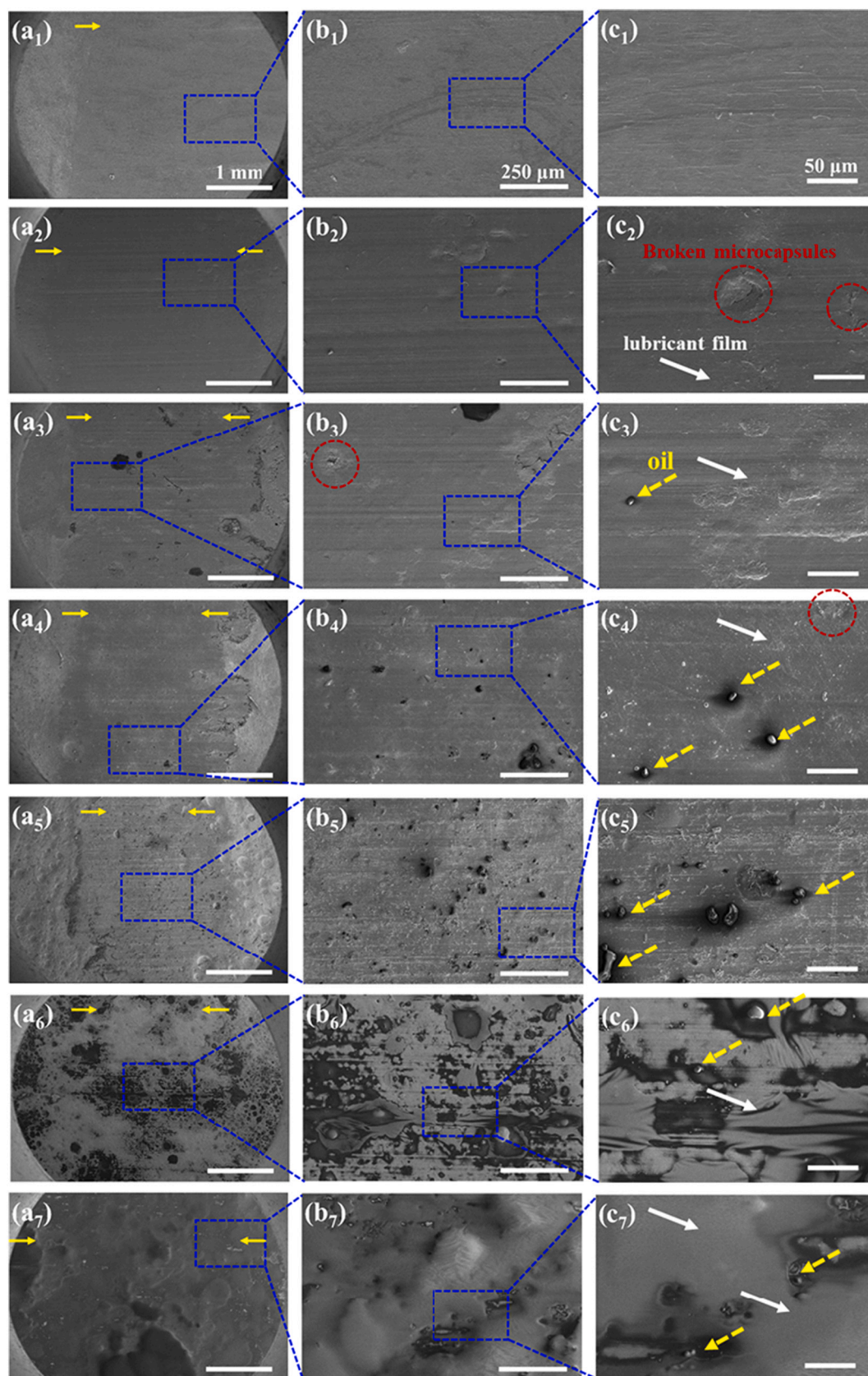


Fig. 6. SEM images of the worn surfaces of C0, C5, C10, C15, C20, C30, C40. The solid yellow arrows (in a_1 – a_7) define the boundaries of wear scars. The dashed yellow arrows (in c_3 – c_7) and the dashed red circles (in b_3 , c_2 , and c_4) correspond to the oil droplets and the broken microcapsules, respectively. The scale bars of a_1 – a_7 , b_1 – b_7 and c_1 – c_7 are 1 mm, 250 μm and 50 μm , respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

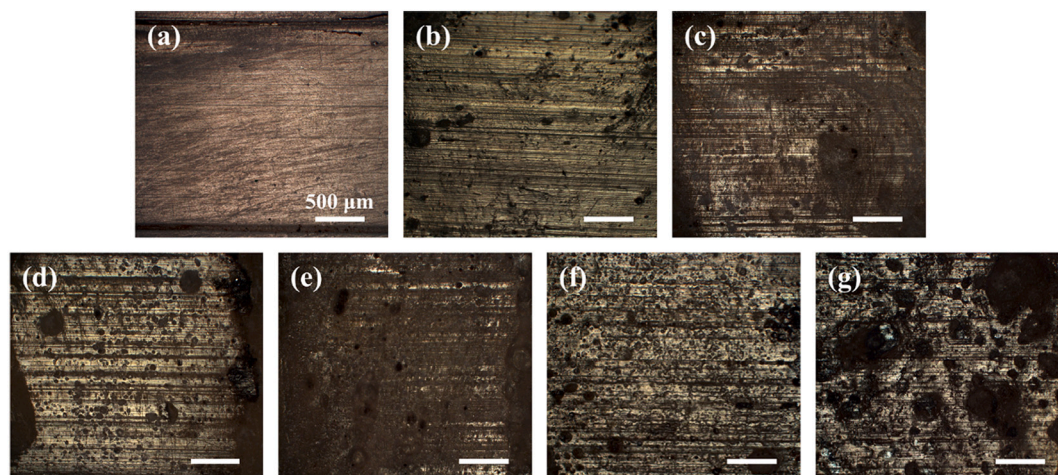


Fig. 7. (a–g) LSCM images of the worn surface of C0, C5, C10, C15, C20, C30, C40. Scale bars in all images are 500 μm .

Table 2

EDS results of the worn surface of MC PA6 self-lubricating coatings with different microcapsule contents.

Sample code	C (wt%)	N (wt%)	O (wt%)	Fe (wt%)	Na (wt%)
C0	68.8	12.4	18.3	0.6	0.0
C5	65.7	11.9	20.7	1.0	0.7
C10	68.3	12.2	17.4	1.9	0.2
C15	66.5	12.9	19.4	0.7	0.6
C20	70.6	12.4	16.5	0.4	0.3
C30	81.6	5.3	11.5	1.0	0.2
C40	99.3	0.0	0.5	0.1	0.0

the microcapsule-loaded MC PA6 composite coating is shown in Fig. 8. When the pre-embedded microcapsules are crushed under mechanical load during the friction processes, the encapsulated lube base oil is released to form effective boundary oil films on the contact surface [53,54]. The formation of boundary films, consequently, changes the interfacial contact status from solid-solid contact to local solid-liquid contact [55], which plays an active role in significantly reducing the friction coefficient and specific wear rate. Besides, part of the broken PUF shells can also adhere to the coating surface as solid lubricants, which combine with the boundary oil film to form lubricating film,

preventing the direct contact between the composite coatings and steel ring [29,48].

A noteworthy discovery is that the worn surfaces of the composite coating do not have obvious holes and interfacial cracks generated from the complete falling-off of microcapsules. This is benefitting from the rough surfaces of PUF microcapsules, which are capable of forming mechanical interlocks with the MC PA6 matrix [56] to prevent the debonding of the PUF shell. In addition, the ruptured microcapsules also play a non-negligible role in trapping wear debris to reduce the friction coefficient and wear rate. As the wear process progresses, wear debris are trapped in the cavities of ruptured microcapsules formed from the release of lube base oil. The entrapment effect of ruptured microcapsules reduces the amount of wear debris, avoiding accumulation of excessive wear debris, thus weakens the abrasive action of the accumulated debris as a third body at the contact area, which further reduces the friction coefficient and wear rate [57,58].

4. Conclusions

In this work, PUF microcapsules containing lube base oil are prepared and introduced into MC PA6 to form a series of dense self-lubricating composite coatings, which obtain remarkable improvement

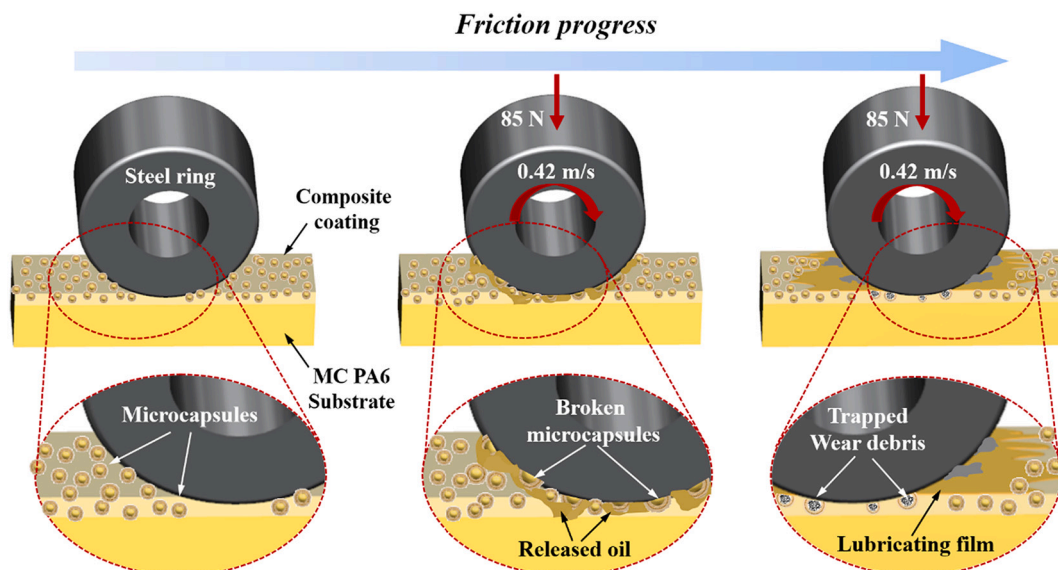


Fig. 8. The schematic diagram of antifriction mechanism for self-lubricating MC PA6 composite coating containing oil-loaded microcapsules.

on the tribological properties. Specially, MC PA6 composite coating incorporating 20 wt% microcapsules displays a friction coefficient of 0.16 and specific wear rate of $0.47 \times 10^{-5} \text{ mm}^3/\text{Nm}$, which are respectively 56.7% and 91.8% lower than those of pure MC PA6 coating, attributed to the formation of a uniform, continuous and thin lubricating film on the worn surface. As the microcapsule content increases to 40 wt %, the friction coefficient continuously drop to an extremely low level of 0.14 due to the formation of thick oil film on the surface, while the specific wear rate increases a bit because excessive microcapsules result in a loose internal structure of the composite coating and cause a sharp deterioration of its hardness to 64 HD. As for temperature near the friction contact area, by introducing oil-loaded microcapsules, the heat generation is well-controlled and the temperature reduces significantly from 93.7 °C to ~40–50 °C, to successfully avoid greater wear loss from thermal softening. Considering the friction coefficient, wear rate, friction temperature, coating hardness and cost, etc., the MC PA6 composite coating with 20 wt% of oil-loaded microcapsules possess the optimal comprehensive properties. This work would provide significant inspiration and guidance for the development and application of MC PA6 composite coatings with good self-lubricating properties that may even be applicable in harsh lubrication conditions.

CRedit authorship contribution statement

Ping-Ping Gao: Methodology, Validation, Formal analysis, Investigation, Data curation, Writing – original draft. **Zhao-Bo Sun:** Investigation, Data curation. **Ya-Jie Mao:** Investigation, Data curation. **Mingwang Pan:** Methodology, Conceptualization. **Xu Ji:** Supervision, Project administration. **Jian-Hua Tang:** Supervision, Project administration. **Kun Dai:** Methodology, Conceptualization. **Hao Lin:** Conceptualization, Resources, Writing – review & editing, Supervision, Project administration, Funding acquisition. **Gan-Ji Zhong:** Methodology, Conceptualization, Resources, Writing – review & editing, Funding acquisition, Supervision, Project administration. **Zhong-Ming Li:** Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Structure and properties of PUF microcapsules containing 32# lube base oil; Tribological performances of MC PA6 self-lubricating composite coatings with different oil viscosity. Supplementary data to this article can be found online at <https://doi.org/10.1016/j.porgcoat.2021.106528>.

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